

AP-E17 Recovery-Compatible Boundary Transfer, Strong-Quasiconvex Dirac Rigidity, and the Nonvolumetric WZ-Capacity Alternative

Route-F derivation ledger

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Abstract

This note executes the three AP-E16 continuation gates and separates the closed statements from the remaining endpoint assumption. On the volumetric part of the relaxation defect, a diagonal minimizing sequence already has vanishing normalized *full* local comparison deficit. The missing operation is not quasiminimality but replacement of its oscillatory boundary trace by the affine barycentric tangent.

We formulate an explicit graph-tight annulus condition and prove a recovery-compatible, sphere-valued boundary modification under precisely that condition. The comparison preserves the original trace, relative homotopy, and global degree, while its normalized energy converges to the affine tangent energy. Combining it with the AP-E12 exact strong-quasiconvexity identity gives an exact variance formula for the homogeneous tangent graph Young measure. Normalized quasiminimality then forces both its ordinary variance and its concentration mass to vanish; the measure is Dirac. Consequently the volumetric defect is zero wherever the graph-tight annulus condition holds.

The condition is not derived from the present endpoint graph bounds. A rank-two microball scaling shows why ordinary $W^{1,2}$ decomposition is insufficient: strong L^2 convergence and bounded first/second-minor energy can coexist with a divergent weighted gluing term. Thus the result closes the Young-measure rigidity implication, but not the endpoint annular replacement theorem.

For every nonvolumetric tangent we prove an independent alternative. The local Wess–Zumino number $\mathbf{b}(r)$ obeys the sharp capacity inequality

$$\mathcal{E}_3(B_r) \geq \frac{3K\pi^3}{2r^3} |\mathbf{b}(r)|^2.$$

Hence a defect of scale r^d has $|\mathbf{b}(r)| = O(r^{(d+3)/2})$ and cannot carry a nonzero quantized local degree. After normalization, either a positive fraction of the tangent is forced by this vanishing but capacity-active WZ flux, or the tangent is WZ-neutral and any remaining obstruction is analytic rather than topological. Hessian, determinant, and portal promotion remain closed.

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1 Frozen graph energy and the volumetric tangent

The AP-E11 derivative density and complete-minor graph map are

$$W(F) = \frac{1}{2}|F|^2 + \frac{R}{2}|M_2(F)|^2 + \frac{K}{2}|M_3(F)|^2, \quad (R, K) = (1, 0.35), \quad (1)$$

$$\Xi(F) = (F, \sqrt{R} M_2(F), \sqrt{K} M_3(F)), \quad W(F) = \frac{1}{2}|\Xi(F)|^2. \quad (2)$$

Let u_j be the smooth degree- B recovery sequence of AP-E15, $\mathcal{E}(u_j) \rightarrow I_B$, with derivative-energy defect

$$W(Du_j) dx \xrightarrow{*} W(Du) dx + \mu. \quad (3)$$

Define the volumetric set

$$\mathcal{V} = \left\{ x_0 : 0 < \vartheta(x_0) = \lim_{r \downarrow 0} \frac{\mu(B_r(x_0))}{|B_r|} < \infty \right\}. \quad (4)$$

On $\mathcal{V}$, μ is absolutely continuous with respect to Lebesgue measure. We may therefore choose x_0 to be simultaneously a density point of (3), a Sobolev differentiability point of u , and a localization point of all complete-minor Young-measure moments. Put

$$q = u(x_0), \quad F_0 = Du(x_0), \quad F_0^T q = 0. \quad (5)$$

Choose $r_\ell \downarrow 0$ and then the diagonal j_ℓ so that

$$\frac{\mathcal{E}(u_{j_\ell}) - I_B}{\mu(B_{r_\ell}(x_0))} \rightarrow 0, \quad (6)$$

$$\frac{1}{r_\ell^5} \int_{B_{r_\ell}(x_0)} |u_{j_\ell} - u|^2 dx \rightarrow 0. \quad (7)$$

The second choice is possible after the radii are fixed because $u_j \rightarrow u$ strongly in L^2 . With $B = B_1(0)$, define the chordal tangent

$$v_\ell(y) = \frac{u_{j_\ell}(x_0 + r_\ell y) - q}{r_\ell}, \quad v_0(y) = F_0 y. \quad (8)$$

Sobolev differentiability and (7) give

$$v_\ell \rightarrow v_0 \quad \text{strongly in } L^2(B). \quad (9)$$

Notice the exact scaling identity $D_y v_\ell = Du_{j_\ell}(x_0 + r_\ell y)$.

2 What full local quasiminimality already gives

For $Q \Subset \Omega$, AP-E16 defines the relative-homotopy local deficit

$$\epsilon_j(Q) = \mathcal{E}(u_j; Q) - \inf\{\mathcal{E}(w; Q) : w = u_j \text{ near } \partial Q, w \simeq u_j \text{ rel. } \partial Q\}. \quad (10)$$

Replacing the map only in Q preserves its exterior boundary value and global degree, hence

$$0 \leq \epsilon_j(Q) \leq \mathcal{E}(u_j) - I_B. \quad (11)$$

Proposition 2.1 (Normalized full local quasiminimality). *For every fixed $0 < \rho < 1$, the diagonal (6) satisfies*

$$\frac{\epsilon_{j_\ell}(B_{\rho r_\ell}(x_0))}{\mu(B_{r_\ell}(x_0))} \rightarrow 0. \quad (12)$$

Proof. Apply (11) and divide by $\mu(B_{r_\ell}(x_0))$. This uses every target-valued, trace-preserving, relative-homotopy comparison, not merely the compact fibre phase. $\square$

Thus AP-E16's desired normalized quasiminimality was already contained in the global diagonal. The actual missing step is to build a comparison whose trace is u_{j_ℓ} but whose interior is the affine tangent.

3 Graph-tight annuli and boundary modification

Let $A_\delta = B \setminus \overline{B_{1-\delta}}$. Choose $\eta_\ell \in C^\infty(B; [0, 1])$ with

$$\eta_\ell = 0 \text{ on } B_{1-2\delta_\ell}, \quad \eta_\ell = 1 \text{ near } \partial B, \quad |D\eta_\ell| \leq C/\delta_\ell. \quad (13)$$

Definition 3.1 (Graph-tight annulus condition, GTA). *The tangent sequence has a graph-tight annulus if there are $\delta_\ell \downarrow 0$ and η_ℓ as above such that*

$$q \cdot u_{j_\ell}(x_0 + r_\ell y) \geq c_0 > 0 \quad \text{on } A_{2\delta_\ell}, \quad (14)$$

$$\int_{A_{2\delta_\ell}} [W(Dv_\ell) + W(F_0)] dy \rightarrow 0, \quad (15)$$

$$\int_{A_{2\delta_\ell}} \frac{|v_\ell - v_0|^2}{\delta_\ell^2} (1 + R|Dv_\ell|^2 + K|M_2(Dv_\ell)|^2) dy \rightarrow 0. \quad (16)$$

Condition (14) is the target-valued trace gate; (15) prevents energy from being trapped on the chosen boundary layer; and (16) is exactly the rank-one cutoff remainder at all three minor orders. No unspecified $o(1)$ is hidden in the definition.

Set

$$\widehat{v}_\ell = \eta_\ell v_\ell + (1 - \eta_\ell) v_0, \quad \widehat{u}_\ell(x_0 + r_\ell y) = \frac{q + r_\ell \widehat{v}_\ell(y)}{|q + r_\ell \widehat{v}_\ell(y)|}. \quad (17)$$

Outside $B_{r_\ell}(x_0)$ leave $\widehat{u}_\ell = u_{j_\ell}$.

Lemma 3.2 (Rank-one graph estimate). *On the transition annulus,*

$$D\widehat{v}_\ell = \eta_\ell Dv_\ell + (1 - \eta_\ell) F_0 + (v_\ell - v_0) \otimes D\eta_\ell. \quad (18)$$

The last term has rank one. Consequently

$$\begin{aligned} W(D\widehat{v}_\ell) \leq C & \left[1 + W(Dv_\ell) + W(F_0) \right. \\ & \left. + \frac{|v_\ell - v_0|^2}{\delta_\ell^2} (1 + R|Dv_\ell|^2 + K|M_2(Dv_\ell)|^2) \right]. \end{aligned} \quad (19)$$

Proof. The first-order estimate follows directly from (18). If $A = \eta_\ell Dv_\ell + (1 - \eta_\ell) F_0$ and $H = (v_\ell - v_0) \otimes D\eta_\ell$, then $\text{rank } H \leq 1$. Multilinearity of determinants gives

$$M_2(A + H) = M_2(A) + L_2(A, H), \quad M_3(A + H) = M_3(A) + L_3(A, A, H),$$

with $|L_2|^2 \leq C|A|^2|H|^2$ and $|L_3|^2 \leq C|M_2(A)|^2|H|^2$. Convex interpolation bounds the minors of A by the corresponding quantities for Dv_ℓ and F_0 . Using $|H| \leq C|v_\ell - v_0|/\delta_\ell$ proves (19). $\square$

Theorem 3.3 (Recovery-compatible boundary modification). *If GTA holds, then (17) is well defined for all sufficiently large ℓ , is sphere valued, equals u_{j_ℓ} near $\partial B_{r_\ell}(x_0)$, and is homotopic to it relative to that boundary. Moreover*

$$\limsup_{\ell \rightarrow \infty} \frac{1}{r_\ell^3} \int_{B_{r_\ell}(x_0)} W(D\widehat{u}_\ell) dx \leq |B|W(F_0). \quad (20)$$

The normalized difference of the potential terms also tends to zero.

Proof. Near the boundary, $\widehat{v}_\ell = v_\ell$, so $q + r_\ell \widehat{v}_\ell = u_{j_\ell}$ and $\widehat{u}_\ell = u_{j_\ell}$ exactly. In the transition layer, (14) and $q + r_\ell v_0 \rightarrow q$ uniformly keep the straight interpolation away from the origin. Radial projection therefore supplies a relative-boundary homotopy and preserves global degree.

For $\mathcal{P}(z) = z/|z|$,

$$D\mathcal{P}(z) = \frac{\text{Id} - \mathcal{P}(z) \otimes \mathcal{P}(z)}{|z|}. \quad (21)$$

After the change of variables $x = x_0 + r_\ell y$,

$$D_x \widehat{u}_\ell = D\mathcal{P}(q + r_\ell \widehat{v}_\ell) D_y \widehat{v}_\ell.$$

On $B_{1-2\delta_\ell}$, $\widehat{v}_\ell = v_0$, and $D_x \widehat{u}_\ell \rightarrow F_0$ uniformly. On the annulus, $D\mathcal{P}$ and its exterior powers are uniformly bounded. Therefore (19) together with (15)–(16) makes the annular energy $o(1)$. This proves (20).

The potential $m^2(1 - u^0)$ is Lipschitz on $\mathbb{S}^3$. Both the affine interior and the GTA transition converge in measure to q , while the annular volume tends to zero. Its normalized comparison error is therefore $o(1)$. $\square$

4 Transfer to the tangent and exact Dirac rigidity

Theorem 4.1 (Volumetric quasiminimality transfer). *At a point $x_0 \in \mathcal{V}$ satisfying GTA,*

$$\limsup_{\ell \rightarrow \infty} \frac{1}{r_\ell^3} \int_{B_{r_\ell}(x_0)} W(Du_{j_\ell}) \, dx \leq |B|W(F_0). \quad (22)$$

Consequently $\vartheta(x_0) = 0$.

Proof. Use $\widehat{u}_\ell$ in (10) and combine (12) with (20). Since $\mu(B_{r_\ell}) \sim \vartheta(x_0)|B|r_\ell^3$, the normalized local deficit is $o(r_\ell^3)$. On the other hand, differentiation of (3) gives

$$\lim_{\ell \rightarrow \infty} \frac{1}{r_\ell^3} \int_{B_{r_\ell}(x_0)} W(Du_{j_\ell}) \, dx = |B|\{W(F_0) + \vartheta(x_0)\}.$$

Comparison with (22) forces $\vartheta(x_0) = 0$. $\square$

The same conclusion has a more informative Young-measure form. Let $(\nu, \lambda, \nu^\infty)$ be the homogeneous generalized tangent Young measure generated by the graph vectors $\Xi(Dv_\ell)$. Here ν is the ordinary oscillation measure, while λ and ν^∞ record quadratic graph concentration. Localization and the null-Lagrangian moment identities give

$$\langle F \rangle_\nu = F_0, \quad \langle M_k(F) \rangle_\nu = M_k(F_0), \quad k = 2, 3. \quad (23)$$

Theorem 4.2 (Exact graph-variance identity and Dirac rigidity). *For the homogeneous tangent,*

$$\begin{aligned} \overline{W} - W(F_0) &= \frac{1}{2} \int [|F - F_0|^2 + R |M_2(F) - M_2(F_0)|^2 \\ &\quad + K |M_3(F) - M_3(F_0)|^2] \, d\nu(F) + \frac{1}{2} \lambda(B). \end{aligned} \quad (24)$$

If normalized quasiminimality is transferred by GTA, then $\overline{W} \leq W(F_0)$ and every term on the right vanishes. Hence

$$\boxed{\nu = \delta_{F_0}, \quad \lambda = 0.} \quad (25)$$

Proof. Expand the three squares in W . The cross terms vanish by (23); the quadratic recession of $\frac{1}{2}|\Xi|^2$ equals $\frac{1}{2}$ on the graph unit sphere and contributes $\lambda(B)/2$. This proves (24). The transfer theorem gives the reverse inequality, whereas (24) is nonnegative. Equality forces $F = F_0$ ν -a.s. and zero concentration mass. $\square$

This is the requested strong-quasiconvex conclusion: once the correct boundary comparison exists, there is no nontrivial homogeneous tangent Young measure left to classify.

5 Why GTA is not yet a consequence of the graph bound

The classical decomposition lemma for an L^p -bounded sequence of gradients can separate an L^p -equiintegrable part from concentration [1]. Generalized gradient Young-measure theory also describes boundary concentration for one fixed p -growth [2]. Neither statement modifies one map while simultaneously preserving the algebraic identities

$$M_2(Dv) = Dv \wedge Dv, \quad M_3(Dv) = Dv \wedge Dv \wedge Dv,$$

their separate endpoint L^2 bounds, the $\mathbb{S}^3$ target, and relative degree.

The missing weighted estimate cannot be recovered from strong L^2 convergence by Hölder alone. Let $\psi \in C_c^\infty(B_1; \mathbb{R}^2)$ have a nonzero second minor, embed it in $\mathbb{R}^4$, and set on a ball of radius h

$$z_h(x) = a_h \psi(x/h), \quad a_h = h^{1/4}. \quad (26)$$

Its derivative has rank at most two, and scaling gives

$$\begin{aligned} \int |z_h|^2 &\asymp h^{7/2}, & \int |Dz_h|^2 &\asymp h^{3/2}, \\ \int |M_2(Dz_h)|^2 &\asymp 1, & M_3(Dz_h) &= 0. \end{aligned} \quad (27)$$

Yet if the cutoff thickness is $\delta = h$,

$$\delta^{-2} \int |z_h|^2 |M_2(Dz_h)|^2 \asymp h^{-3/2} \longrightarrow \infty. \quad (28)$$

Radial projection of $(1, z_h)$ to $\mathbb{S}^3$ preserves these leading powers. This example does not prove that every annulus is bad; it proves the precise logical point that the current strong L^2 and complete-minor energy bounds do not imply (16). A complete-minor equiintegrability/trace theorem, or a model-specific reverse-Hölder estimate, is still required.

6 Nonvolumetric concentration and WZ capacity

For a smooth $u : B_r \rightarrow \mathbb{S}^3$, write

$$J_u = \det[u, \partial_1 u, \partial_2 u, \partial_3 u], \quad \mathbf{b}_u(r) = \frac{1}{2\pi^2} \int_{B_r} J_u \, dx. \quad (29)$$

With the AP-E14 orientation, $\mathbf{b}$ is also $(16\pi^2)^{-1} \int a \wedge da$. Pointwise $|J_u| \leq |M_3(Du)|$.

Theorem 6.1 (Sharp local WZ-capacity lower bound). *For every ball,*

$$\boxed{\mathcal{E}_3(u; B_r) = \frac{K}{2} \int_{B_r} |M_3(Du)|^2 \, dx \geq \frac{3K\pi^3}{2r^3} |\mathbf{b}_u(r)|^2.} \quad (30)$$

In Hopf variables the first/curvature components also obey

$$\left(\frac{1}{8} \int_{B_r} |a|^2 \right) \left(\frac{R}{32} \int_{B_r} |da|^2 \right) \geq R\pi^4 |\mathbf{b}_u(r)|^2. \quad (31)$$

Proof. Cauchy–Schwarz and $|B_r| = 4\pi r^3/3$ give

$$(2\pi^2 |\mathbf{b}|)^2 \leq |B_r| \int_{B_r} |M_3(Du)|^2.$$

Multiplication by $K/2$ proves (30). For (31), use $(16\pi^2 \mathbf{b})^2 \leq (\int |a|^2)(\int |da|^2)$. $\square$

Corollary 6.2 (No quantized point-degree concentration). *Suppose along a tangent sequence*

$$\mathcal{E}(u_{j_\ell}; B_{r_\ell}(x_0)) \leq Cr_\ell^d \quad (d \geq 0). \quad (32)$$

Then

$$|\mathbf{b}_{u_{j_\ell}}(r_\ell)| \leq \left(\frac{2C}{3K\pi^3} \right)^{1/2} r_\ell^{(d+3)/2} \longrightarrow 0. \quad (33)$$

If the boundary is constant so that $\mathbf{b}$ is an integer relative degree, it is therefore zero for all sufficiently large ℓ .

Thus no nonvolumetric defect can hide a nonzero integer $B = 1$ bubble at a point. The Hopf base has no separate monopole charge either: since a is a globally defined one-form on the ball,

$$\int_{\partial B_r} da = \int_{B_r} d(da) = 0. \quad (34)$$

For rational $0 < \rho < 1$, define the normalized capacity fraction

$$\kappa_\ell(\rho) = \frac{3K\pi^3}{2} \frac{|\mathbf{b}_{u_{j_\ell}}(\rho r_\ell)|^2}{(\rho r_\ell)^3 \mu(B_{r_\ell}(x_0))}. \quad (35)$$

After one diagonal subsequence, every $\kappa_\ell(\rho)$ has a limit. Equation (30) gives the following exhaustive alternative.

Theorem 6.3 (Nonvolumetric capacity/topology alternative). *At every nonvolumetric tangent point, exactly one of the following occurs.*

1. **Capacity-active branch.** *For some rational ρ , $\lim \kappa_\ell(\rho) > 0$. A positive fraction of the normalized tangent energy is forced into the sextic/WZ component. Its absolute WZ number still tends to zero by (33), so it is a fractional capacity effect, not a quantized topological atom.*
2. **WZ-neutral branch.** *For every rational ρ , $\kappa_\ell(\rho) \rightarrow 0$. The tangent carries neither an integer local degree nor a positive normalized WZ-capacity fraction. Any surviving defect is analytically concentrated in the graph variables and is not protected by Route-E topology.*

The alternative is a classification, not yet an exclusion theorem. The capacity-active branch must be retained as an order-specific concentration coordinate. The WZ-neutral branch can be removed only after a GTA-type trace theorem is proved for generalized graph tangents.

7 Gate decision

The three requested tasks now have the following exact status.

1. **Boundary transfer: conditional theorem proved.** Normalized full local quasiminimality is unconditional on the AP-E16 diagonal. Its transfer to the affine tangent is proved under GTA. GTA itself is not implied by the present endpoint graph bounds.
2. **Dirac rigidity: proved once transfer holds.** Equation (24) forces both oscillation variance and concentration mass to vanish, so the homogeneous tangent is exactly δ_{F_0} .
3. **Nonvolumetric alternative: proved unconditionally.** Equations (30)–(35) exclude quantized point-degree concentration and split every remaining tangent into the capacity-active or WZ-neutral branch.

The new mainline blocker is therefore narrower than AP-E16: prove GTA for the actual minimizing diagonal, most plausibly by a complete-minor equiintegrability theorem or a model-specific reverse-Hölder estimate. Until that endpoint trace theorem closes, full $\mu = 0$, local recovery, classicality, isolation, and the bosonic Hessian remain false. Determinant and degree-one portal promotion also remain false.

Reproducibility. Run `python3 route_f/code/verify_ap_e17_boundary_transfer_dirac_capacity.py` from the repository root. The card verifies radial projection, exact strong-quasiconvex variance, the microball scaling obstruction, WZ-capacity constants, the nonvolumetric exponent, and every fail-closed gate. It performs no lattice relaxation or grid scan.

References

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